

V9.03 Multi-Seed Configuration

Multi-seed configuration

```
RUN_LABEL           = L=16, a=21.0, alpha=74.2, beta_init=0.0
RUN_PRESET          = size_default
L_NM, A_NM, ALPHA_DEG = 16.0, 21.0, 74.2
BETA_INIT_DEG       = 0.0
ANIS_MODEL          = cubic_first_raw
DIPOLE_INIT_MODE     = random
MOVE_POSITIONS      = True
MOVE_ORIENTATIONS   = True
GLOBAL_COTILT        = 1 moves/cycle, 0.75 deg
N_MAG_PER_CYCLE      = 1
TRANS_STEP_NM        = 0.03
ROT_STEP_DEG         = 2.0
DIP_STEP_RAD         = 0.3
NN_SHELL_FACTOR      = 1.18
B_FIELD_T            = 0.05
N_VOXEL              = 4
```

```
MULTI_SEEDS          = [1, 2, 3, 4, 5]
MULTI_CYCLES, MULTI_EQUIL = 10000, 2000
MULTI_RESEED_INITIAL_STATE = True
```

Each per-seed row is the equilibrated mean over cycles MULTI_EQUIL..MULTI_CYCLES-1.

V9.03 Multi-Seed Per-Seed Results: Tilt and energy

seed	elapsed_min	E_mean	local_beta	body_tilt	SL_PCA	SL_NN	body_SL_mismatch
1	22.9811	-627.371	9.17629	4.36198	4.43593	5.45703	1.64258
2	23.2911	-625.535	9.65418	4.82801	4.81793	6.27073	1.46755
3	22.6074	-626.077	9.37488	5.43613	5.58864	5.99434	1.13716
4	22.8521	-623.043	9.89884	4.48222	4.08366	4.56805	1.84633
5	22.5888	-625.902	9.64669	5.44814	5.58826	5.98898	1.09794

V9.03 Multi-Seed Per-Seed Results: Order and gaps

seed	body_order	gap_min	gap_p05	muB
1	0.986704	1.81133	2.04519	0.956533
2	0.985528	1.81138	2.04947	0.956637
3	0.988292	1.81213	2.06032	0.957419
4	0.979746	1.81375	2.09618	0.956797
5	0.987033	1.8114	2.05095	0.95758

V9.03 Multi-Seed Per-Seed Results: Acceptance

seed	acc_mech	acc_cotilt	acc_dip
1	0.753919	0.9443	0.723181
2	0.764512	0.9479	0.72463
3	0.767931	0.9401	0.722796
4	0.784458	0.9459	0.722852
5	0.766015	0.9427	0.721752

V9.03 Multi-Seed Mean and Std: Tilt and energy

stat	elapsed_min	E_mean	local_beta	body_tilt	SL_PCA	SL_NN	body_SL_mismatch
mean	22.8641	-625.585	9.55018	4.9113	4.90288	5.65583	1.43831
std	0.290628	1.58077	0.279383	0.513918	0.677565	0.675734	0.322337

V9.03 Multi-Seed Mean and Std: Order and gaps

stat	body_order	gap_min	gap_p05	muB
mean	0.98546	1.812	2.06042	0.956993
std	0.00334288	0.00103266	0.0207399	0.000474919

V9.03 Multi-Seed Mean and Std: Acceptance

stat	acc_mech	acc_cotilt	acc_dip
mean	0.767367	0.94418	0.723042
std	0.0109913	0.0029853	0.00103695

multi-seed coherent body and SL tilt

